Unrolling Plug-and-Play Network for Hyperspectral Unmixing

Min Zhao, *Graduate Student Member, IEEE*, Linruize Tang, *Graduate Student Member, IEEE*, and Jie Chen, *Senior Member, IEEE*

***Abstract*— Deep learning-based unmixing methods have received great attention in recent years and achieved remarkable performance. These methods employ a data-driven approach to extract structure features from hyperspectral images; however, they tend to be less physically interpretable. Conventional unmixing methods have much more interpretability, whereas they require manually designing regularization and choosing penalty parameters. To overcome these limitations, we propose a novel unmixing method by unrolling the plug-and-play unmixing algorithm to conduct the deep architecture. Our method integrates both inner and outer priors. The carefully designed unfolding deep architecture is used to learn the spectral and spatial information from the hyperspectral image, which we refer to as inner priors. Additionally, our approach incorporates deep denoisers that have been pretrained on a large volume of image data to leverage the outer priors. Second, we design a dynamic convolution to model the multiscale information. Different scales are fused with an attention module. Experimental results of both synthetic and real datasets demonstrate that our method outperforms compared methods.**

***Index Terms*— Alternative direction method of multipliers (ADMM), hyperspectral unmixing, inner priors, outer priors, plug-and-play, unrolling.**

I. Introduction

HYPERSPECTRAL imaging stands as a pivotal domain in remote sensing, capturing data that incorporates both spatial and spectral characteristics of the target. The dense spectral information with hundreds of spectral bands allows it to be widely applied in many fields, such as environmental and agriculture monitoring [1], [2], [3]. However, due to the high work distance and low spatial resolution of hyperspectral sensors, a pixel may contain several materials, which degrades the performance of subsequent high-level data processing. Spectral unmixing is one of the most prominent tools to cope with this issue, aiming to decompose the mixed pixels into pure components, termed endmembers, and their corresponding fraction abundances.

Conventional model-based unmixing methods exploit convex optimization algorithms to solve the unmixing objective function with some hand-crafted regularizations based on the spatial and spectral priors in hyperspectral images. For instance, in most scenes, the pixel values of an image are piecewise continuous in the spatial dimension. Iordache et al. [4], Xiong et al. [5], and Li et al. [6] introduce total variation (TV) to achieve this purpose. As a pixel usually contains much fewer materials than the number of pure materials contained in the endmember matrix, sparse constraints are introduced to the abundance matrix to obtain sparse results [4], [6]. Minimum-volume simplex regularization is incorporated in the objective function to constrain the volume of the simplex formed by endmembers [7].

Typically, the alternative direction method of multipliers (ADMM) is a powerful tool to solve unmixing problems, which decomposes the complicated optimization problem into several easier subproblems and iteratively solves them. Iordache et al. [4] utilizes this strategy to address the objective function with two regularizations on abundance, namely ℓ_1-norm and TV constraints. A nonstandard application of ADMM with a block coordinate descent scheme is designed to address a 3-D TV-constrained problem, which can model the spatial and spectral correlations and gain sharper edges [8]. In [9], an ADMM-based joint endmember and abundance estimation hyperspectral unmixing method is proposed, which can simultaneously estimate abundances and endmembers. A graph TV regularization is considered to capture the spatial correlation information. Li et al. [10] use this tool to solve a graph regularized nonlinear unmixing method with the multilinear mixing model to indicate the nonlinearity between endmembers. Many other ADMM-based unmixing methods have also been proposed, including those for nonlinear unmixing using kernel methods [11] and those addressing unmixing problems that consider spectral variability [12]. While ADMM is a versatile and powerful tool for solving unmixing optimization problems, the choice of penalty parameters inherently limits performance.

Compared to traditional methods that rely on manually designed regularization terms, deep neural networks (DNNs) can automatically extract and model complex structure patterns and nonlinear relationships in data. DNNs have achieved breakthrough results in hyperspectral image processing tasks such as object recognition [13], band selection [14], and image super-resolution [15]. The encoding and decoding processes of an autoencoder perfectly fit the formulation of unmixing problem, and these kinds of methods have achieved great advances [16]. Many unmixing methods have been

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The authors are with the School of Marine Science and Technology, Northwestern Polytechnical University, Xi'an 710072, China, and also with the Research and Development Institute, Northwestern Polytechnical University, Shenzhen 518063, China (e-mail: minzhao@mail.nwpu.edu.cn; lrztang@mail.nwpu.edu.cn; dr.jie.chen@ieee.org).

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proposed based on deep autoencoders [17]. To effectively leverage the spatial structure information of hyperspectral images, deep convolutional neural network (CNN)-based autoencoders are conducted [18], [19]. Some unmixing networks with modified autoencoder architectures have also been proposed. For instance, MSSS-Net [20] builds a two-stream unmixing framework, which adopts an end-to-end manner to simultaneously learn spatial stream and multiview spectral stream networks, and fuses information of different scales to achieve more effective unmixing. In work [21], an endmember-guided subnetwork is introduced alongside the basic autoencoder framework to extract features from pure or near-pure spectra. Meanwhile, a weight-sharing strategy is used to guide the learning of basic network. However, DNNs are often regarded as "black boxes" and lack physical interpretability.

Recently, integration model- and deep learning-based methods is a new trend in designing unmixing approaches with both interpretability and data-driven advantages. On one hand, the plug-and-play unmixing approaches use a pretrained denoiser to provide priors [22], [23]. However, these methods still require setting the value of penalty parameters. On the other hand, unrolling/unfolding the optimization algorithm to design the unmixing network structure receives great performance [24], [25], [26], [27], which breaks through the drawbacks of iterative algorithms and plugs interpretability to the deep architecture. Zhou and Rodrigues [26] unroll the constrained sparse regression (CSR) problem to construct the abundance estimation network, which has shown better performance with lighter structures and faster convergence speed. Kong et al. [27] employ a full CNN to unroll the variable splitting and augmented Lagrangian algorithm. However, the spatial information learned by this model is still unsatisfactory, and its performance is limited in high-noise scenes.

In our work, to overcome the poor interpretability of deep learning-based unmixing architecture, we propose a novel unmixing network, named PnP-Net. Our method unrolls the plug-and-play framework into a deep network scheme. It allows the optimization-based unmixing model to adapt to data-dependent structures by learning optimal parameters through training on data. Plugging pretrained deep denoiser can effectively leverage the information from additional data and introduce interpretability to the network. The main merits of this work are threefold.

1) We propose a novel plug-and-play unrolling framework for hyperspectral unmixing, where we unroll the ADMM-based plug-and-play algorithm and design the network layers to mimic the iterative process. To the best of our knowledge, our method is the first attempt to combine the plug-and-play priors with deep unfolding in hyperspectral unmixing, which takes advantage of both the optimization- and learning-based methods.
2) Our proposed framework can easily plug various deep denoisers in the network architecture bypassing the complicated network design, which is highly scalable and efficient. Based on the plugged denoisers, our proposed framework leverages the information from additional data and combines the priors with internal learning from the hyperspectral data. It also enhances the interpretability of our network architecture.
3) In this work, we design a dynamic convolution using kernels of various sizes to capture multiscale features for spectral unmixing. It increases the model complexity with multiple parallel convolution kernels fused by an attention module.

The remainder of this article is organized as follows. We describe the background of our method in Section II. In Section III, we present our proposed PnP-Net method in detail and give the main flowchart and training details. Section IV shows the experiment results to validate the effectiveness of our method. Section V concludes this work.

II. Background

In this section, we briefly review the basic concepts of the plug-and-play unmixing framework and deep unrolling network.

From the standpoint of physical interpretability and simplicity, the most commonly used mixing model is the linear mixing model (LMM), presuming that an observed pixel is a linear combination of endmembers weighted by abundances, i.e.,

$$\mathbf{x} = \mathbf{M}\mathbf{a} + \mathbf{n} \tag{1}$$

where $\mathbf{x} \in \mathbb{R}^{B\times 1}$ is a vector containing B spectral bands associated with the obtained pixel, $\mathbf{M} \in \mathbb{R}^{B\times R}$ is the endmember matrix with R spectral signatures of pure constituents, $\mathbf{a} \in \mathbb{R}^{R\times 1}$ is the corresponding abundance, and $\mathbf{n}$ denotes the additive noise. Due to the physical interpretation of hyperspectral image and mixing model, the endmembers and abundances are considered to satisfy the nonnegativity constraint (ENC and ANC), and the abundances are also assumed to satisfy sum-to-one for each pixel (ASC). It is important to highlight that the ASC can be ignored for some physically motivated reasons. This may occur, for instance, when there are local variations in the topography of the scene [28]. In our work, we take into account all these constraints, i.e.,

$$\mathcal{D}_M := \{\mathbf{M}|\mathbf{M} \succeq \mathbf{0}\} \tag{2}$$

$$\mathcal{D}_A := \left\{\mathbf{a}|\mathbf{a} \succeq \mathbf{0}, \sum_{i=1}^{R} a_i = 1\right\} \tag{3}$$

in which $\mathcal{D}_M$ and $\mathcal{D}_A$ are the feasible regions of endmembers and abundances, respectively.

For a hyperspectral image $\mathbf{X} \in \mathbb{R}^{B\times N}$ with N pixels, the hyperspectral unmixing task is formulated by the following optimization problem:

$$\min_{\substack{\mathbf{M}\in\mathcal{D}_M \\ \mathbf{A}\in\mathcal{D}_A}} \frac{1}{2}\|\mathbf{X} - \mathbf{M}\mathbf{A}\|_{\mathrm{F}}^2 + \lambda\mathcal{R}(\mathbf{M}, \mathbf{A}) \tag{4}$$

where the first term is the data fitting term, the second term serves as regularization aiming to enforce certain desirable properties of endmembers and abundances, and λ is the tradeoff parameter.

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A. Plug-and-Play Unmixing Methods

The plug-and-play framework benefits from the variable splitting technique, allowing the utilization of denoising priors to effectively tackle a range of image restoration problems. It can incorporate pretrained denoising networks, with powerful prior modeling capabilities, into the iterative optimization process and has been used for hyperspectral unmixing. Based on the LMM, the work [29] embeds various denoisers for abundance estimation with known endmembers. Zhao et al. [30] combine the sparsity constraint and denoiser prior to solving the joint endmember and abundance estimation-based unmixing problem. In [22], the denoising regularization is introduced in the nonlinear unmixing. Typically, the denoising regularization is plugged for the abundances with (4) rewritten as

$$\min_{\substack{\mathbf{M}\in\mathcal{D}_M\\ \mathbf{A}\in\mathcal{D}_A}} \frac{1}{2}\|\mathbf{X}-\mathbf{M}\mathbf{A}\|_{\mathrm{F}}^2+\lambda\mathcal{R}(\mathbf{A}). \tag{5}$$

We use ADMM to resolve the optimization problem described in (5). By introducing an auxiliary variable $\mathbf{V}$ to replace $\mathbf{A}$ in the regularization term of (5) and adding constraint $\mathbf{V}=\mathbf{A}$, the original problem is decomposed into easier and more manageable subproblems. The iterative optimization process of (5) is as follows:

$$\{\mathbf{M},\mathbf{A}\}\leftarrow \arg\min_{\substack{\mathbf{M}\in\mathcal{D}_M\\ \mathbf{A}\in\mathcal{D}_A}} \|\mathbf{X}-\mathbf{M}\mathbf{A}\|_{\mathrm{F}}^2+\frac{\rho}{2}\|\mathbf{A}-\mathbf{V}+\mathbf{G}\|_{\mathrm{F}}^2 \tag{6}$$

$$\mathbf{V}\leftarrow \arg\min_{\mathbf{V}} \frac{\rho}{2}\|\mathbf{A}-\mathbf{V}+\mathbf{G}\|_{\mathrm{F}}^2+\lambda\mathcal{R}(\mathbf{V}) \tag{7}$$

$$\mathbf{G}\leftarrow \mathbf{G}+\mathbf{A}-\mathbf{V} \tag{8}$$

in which $\mathbf{G}$ is the dual variable, and ρ is the penalty parameter. The process involves two essential operators. The first is a joint endmember and abundance estimation-based unmixing operator to estimate endmembers and abundances. The second step can be viewed as a denoising of $\mathbf{A}+\mathbf{G}$. The regularization term $\mathcal{R}(\mathbf{V})$ can be implicitly coped by incorporating a denoising operator ($\mathsf{C}(\cdot)$), i.e.,

$$\mathbf{V}=\mathsf{C}(\mathbf{A}+\mathbf{G}). \tag{9}$$

In general, the denoiser $\mathsf{C}(\cdot)$ can be any readily available denoising operator. For example, the well-known nonlocal means denoising (NLM) [31] and block-matching and 3-D (BM3D) filtering [32] have been plugged to capture image priors and get high-quality unmixing results. This provides the possibility of integrating CNN-based denoisers with robust prior learning from mass image data and addressing the drawback of insufficient volume of hyperspectral data. The plug-and-play based unmixing methods significantly improve the unmixing performance and provide an effect manner to integrate physical model and deep methods. However, these methods suffer form the limitations of manual parameter setting.

B. Deep Unrolling Network

The deep unrolling paradigm involves unfolding iterative optimization algorithms into trainable deep architectures. One popular deep unrolling unmixing method is unfolding the CSR for sparse unmixing [26]. With known $\mathbf{M}$ and $\mathcal{R}(\mathbf{A})=\|\mathbf{A}\|_1$, it introduces an auxiliary variable $\mathbf{V}=\mathbf{A}$. The ADMM iteratively addresses the optimization problem through the following steps:

$$\mathbf{A}^{(k+1)}=\mathbf{W}\mathbf{X}+\mathbf{Q}\left(\mathbf{V}^{(k)}-\mathbf{G}^{(k)}\right) \tag{10}$$

$$\mathbf{V}^{(k+1)}=\max\left(\mathrm{soft}\left(\mathbf{A}^{(k+1)}+\mathbf{G}^{(k)},\frac{\lambda}{\rho}\right),\mathbf{0}\right) \tag{11}$$

$$\mathbf{G}^{(k+1)}=\mathbf{G}^{(k)}+\mathbf{A}^{(k+1)}-\mathbf{V}^{(k+1)} \tag{12}$$

in which $\mathbf{W}=(\mathbf{M}^\top\mathbf{M}+\rho\mathbf{I})^{-1}\mathbf{M}^\top$ and $\mathbf{Q}=(\mathbf{M}^\top\mathbf{M}+\rho\mathbf{I})^{-1}\rho$. $\mathrm{Soft}(a,\theta)=\mathrm{sign}(a)(|a|-\theta)_+$ is the soft-threshold operator to resolve the ℓ_1-norm. The iterative steps from (10)–(12) can be unfolded into network layers, denoted as Layer_A, Layer_V, and Layer_G with learnable parameters $\{\mathbf{W}^{(k+1)},\mathbf{Q}^{(k+1)},\theta^{(k+1)}\}$, where $\theta^{(k+1)}$ replaces the role of (λ/ρ). Thus, (10)–(12) can be reexpressed as follows:

$$\begin{aligned}\mathbf{A}^{(k+1)}&=\mathrm{Layer}_A\left(\mathbf{V}^{(k)},\mathbf{G}^{(k)};\mathbf{W}^{(k+1)},\mathbf{Q}^{(k+1)}\right)\\&=\mathbf{W}^{(k+1)}\mathbf{X}+\mathbf{Q}^{(k+1)}\left(\mathbf{V}^{(k)}-\mathbf{G}^{(k)}\right)\end{aligned} \tag{13}$$

$$\begin{aligned}\mathbf{V}^{(k+1)}&=\mathrm{Layer}_V\left(\mathbf{A}^{(k+1)},\mathbf{G}^{(k)};\theta^{(k+1)}\right)\\&=\mathrm{ReLU}\left(\mathbf{A}^{(k+1)}+\mathbf{G}^{(k)}-\theta^{(k+1)}\mathbf{I}\right)\end{aligned} \tag{14}$$

$$\begin{aligned}\mathbf{G}^{(k+1)}&=\mathrm{Layer}_G\left(\mathbf{A}^{(k+1)},\mathbf{V}^{(k+1)},\mathbf{G}^{(k)}\right)\\&=\mathbf{G}^{(k)}+\mathbf{A}^{(k+1)}-\mathbf{V}^{(k+1)}.\end{aligned} \tag{15}$$

Each iteration is decomposed into a single network layer, and combining these layers many times is analogous to performing multiple iterations of CSR. The unrolling deep network can be trained in an end-to-end manner. In this way, the knowledge of classical iterative algorithms is infused in the design of deep networks for hyperspectral unmixing. However, the networks obtained by unrolling iterative algorithms are restricted to one pixel only and without considering spatial structure information. By some modifications of networks, we can employ well-designed layers such as convolutional networks to add more spatial priors.

III. Proposed Method

A. Problem Formulation

In (4), handcrafting an effective regularizer $\mathcal{R}(\mathbf{M},\mathbf{A})$ and devising an efficient algorithm to solve the objective function are challenging tasks. Rather than using this intricate path, we propose to derive priors from various image data and integrate them into the optimization process by a plug-and-play strategy, where the regularization is implicitly designed by the utilization of a denoising algorithm. In the hyperspectral unmixing task, the spatial information is embedded in the abundance, and the spectral information is contained in the endmember. Thus, we plug regularization on abundance to fully exploit the spatial information. We introduce regularization by denoising (RED) as the regularizer to add denoising priors, and the corresponding hyperspectral unmixing optimization problem can be reformulated as

$$\min_{\mathbf{M},\mathbf{A}} \frac{1}{2}\|\mathbf{X}-\mathbf{M}\mathbf{A}\|_{\mathrm{F}}^2+\lambda\mathcal{R}_{\mathrm{RED}}(\mathbf{A})+\Omega_{\mathcal{D}_A}(\mathbf{A})+\Omega_{\mathcal{D}_M}(\mathbf{M}) \tag{16}$$

where $\Omega_{\mathcal{D}}$ is an indicator function of a set $\mathcal{D}$ and is defined as

$$\Omega_{\mathcal{D}}(\mathbf{Z})=\begin{cases}0, & \mathbf{Z}\in\mathcal{D}\\ \infty, & \text{otherwise.}\end{cases} \tag{17}$$

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$\mathcal{R}_{\text{RED}}(\mathbf{A})$ is defined as

$$\mathcal{R}_{\text{RED}}(\mathbf{A}) = \frac{1}{2}\mathbf{A}^{\top}\big(\mathbf{A} - \mathsf{C}(\mathbf{A})\big) \tag{18}$$

where $\mathsf{C}(\cdot)$ is an off-the-shelf denoiser. RED depends on the inner product between the solution $\mathbf{A}$ and its residual after denoising $\mathbf{A} - \mathsf{C}(\mathbf{A})$. Under mild assumptions, it demonstrates advantageous derivative properties with $\nabla\mathcal{R}(\mathbf{A}) = \mathbf{A} - \mathsf{C}(\mathbf{A})$. This formulation also provides flexibility in accommodating any denoising engines like the original plug-and-play denoising framework.

We use ADMM algorithm to solve (16). Two auxiliary variables $\mathbf{V}_1$ and $\mathbf{V}_2$ are introduced, and the objective function is rewritten as

$$\begin{aligned}&\min_{\substack{\mathbf{M},\mathbf{A},\\ \mathbf{V}_1,\mathbf{V}_2}} \frac{1}{2}\|\mathbf{X} - \mathbf{M}\mathbf{A}\|_{\text{F}}^2 + \lambda\mathcal{R}_{\text{RED}}(\mathbf{V}_1) + \Omega_{\mathcal{D}_A}(\mathbf{A}) + \Omega_{\mathcal{D}_M}(\mathbf{V}_2)\\ &\text{s.t. } \mathbf{A} = \mathbf{V}_1, \quad \mathbf{M} = \mathbf{V}_2.\end{aligned} \tag{19}$$

The associated augmented Lagrangian is expressed as

$$\begin{aligned}\mathcal{J} = &\frac{1}{2}\|\mathbf{X} - \mathbf{M}\mathbf{A}\|_{\text{F}}^2 + \lambda\mathcal{R}_{\text{RED}}(\mathbf{V}_1) + \Omega_{\mathcal{D}_A}(\mathbf{A}) + \Omega_{\mathcal{D}_M}(\mathbf{V}_2)\\ &+ \frac{\alpha}{2}\|\mathbf{A} - \mathbf{V}_1 + \mathbf{G}_1\|_{\text{F}}^2 + \frac{\beta}{2}\|\mathbf{M} - \mathbf{V}_2 + \mathbf{G}_2\|_{\text{F}}^2\end{aligned} \tag{20}$$

where $\mathbf{G}_1$ and $\mathbf{G}_2$ are dual variables, and α and β are nonnegative parameters. Then the ADMM algorithm involves solving four subproblems at each iteration. In our work, we unroll the ADMM algorithm into a deep network by associating each iteration with a single network layer and then stacking a finite number of layers together. Through unrolling, we can use backpropagation to learn regularization parameters that are difficult to handcraft. This approach allows us to capture image priors through a well-designed network structure and add physical interpretability to the deep network. The proposed unrolling layers (PnP-Net) are presented in detail as follows.

B. Unrolling Plug-and-Play Framework for a Layer

1) Update $\mathbf{A}$*:* The $\Omega_{\mathcal{D}_A}$ can be satisfied with a Softmax operator. The remained part of A-subproblem is a least-square problem defined as

$$\min_{\mathbf{A}} \frac{1}{2}\|\mathbf{X} - \mathbf{M}\mathbf{A}\|_{\text{F}}^2 + \frac{\alpha}{2}\|\mathbf{A} - \mathbf{V}_1 + \mathbf{G}_1\|_{\text{F}}^2. \tag{21}$$

It can be solved with a closed-form as follows:

$$\mathbf{A}^{(k+1)} = \big(\mathbf{M}^{\top}\mathbf{M} + \alpha\mathbf{I}\big)^{-1}\big[\mathbf{M}^{\top}\mathbf{X} + \alpha(\mathbf{V}_1 - \mathbf{G}_1)\big]. \tag{22}$$

The A-update layer is designed by unfolding (22), and it can be rewritten as

$$\begin{aligned}\mathbf{A}^{(k+1)} &= \text{Layer}_A\Big(\mathbf{V}_1^{(k)}, \mathbf{G}_1^{(k)}, \mathbf{X}, \mathbf{W}_1, \mathbf{Q}_1\Big)\\ &= \mathbf{W}_1\mathbf{X} + \mathbf{Q}_1\Big(\mathbf{V}_1^{(k)} - \mathbf{G}_1^{(k)}\Big)\end{aligned} \tag{23}$$

where

$$\begin{aligned}\mathbf{W}_1 &= \big(\mathbf{M}^{\top}\mathbf{M} + \alpha\mathbf{I}\big)^{-1}\mathbf{M}^{\top}\\ \mathbf{Q}_1 &= \big(\mathbf{M}^{\top}\mathbf{M} + \alpha\mathbf{I}\big)^{-1}\alpha.\end{aligned} \tag{24}$$

As for a closed-form solution, conventional unrolling methods only learn the regularization parameters. In [26] and [27], $\mathbf{W}_1$ and $\mathbf{Q}_1$ are considered as the learnable parameters to enhance the flexibility of the network and improve the performance. Following this strategy, we also replace the fixed $\mathbf{W}_1$ and $\mathbf{Q}_1$ with parameters to be estimated.

In [26], two fully connected layers (FCLs) with the bias set to $\mathbf{0}$ adding together are used to formulate Layer_A. In [27], 2-D convolutional layers are used to generate this layer, which can capture spatial information. In our work, inspired by the dynamic convolution work [33], we design a novel dynamic convolution assembling multiscale convolution kernels to formulate this layer, which can capture abundant spatial information from different scales. This layer consolidates multiple parallel convolution kernels dynamically, adjusting their contributions based on input-dependent attention. Instead of enhancing either the depth or the width of the network, this layer possesses increased representational capability due to the nonlinear aggregation of these kernels through attention. Convolutional kernels with different sizes are able to capture multiscale information. We denote the traditional 2-D convolution layer as

$$\mathbf{O} = \mathbf{K} \otimes \mathbf{I}\mathbf{x} \tag{25}$$

where $\mathbf{K}$ is the convolutional kernel, $\mathbf{I}\mathbf{x}$ denotes the input, and $\mathbf{O}$ is the output. Our dynamic convolution layer (DCL) with L parallel multiscale convolution kernels is defined as

$$\mathbf{O} = \mathbf{K}(\mathbf{I}\mathbf{x}) \otimes \mathbf{I}\mathbf{x} \tag{26}$$

where $\mathbf{K}(\mathbf{I}\mathbf{x}) = \sum_{l=1}^{L}(\mathcal{T}[\mathbf{T}_l(\mathbf{I}\mathbf{x})] \odot \mathbf{K}_l)$, $\mathbf{K}_l$ is the convolution kernels with different scales. $\mathbf{T}_l$ is the attention weight of the lth convolution layer, and the size of $\mathbf{T}_l$ is the same as the maximum size of convolutional kernels. The size of available weights of $\mathbf{T}_l$ is the same as the lth kernels, and they are designed on the center of $\mathbf{T}_l$. Other positions are padded with zeros. The summation of each position of $\mathbf{T}$ is one, i.e., $\sum_{l=1}^{L}\mathbf{T}_l(i, j) = 1$ (the (i, j)th position). The size of $\mathbf{T}_l$ is inconsistent with $\mathbf{K}_l$. We use $\mathcal{T}$ to extract the available weights from $\mathbf{T}_l$ to make $\mathbf{T}_l$ and $\mathbf{K}_l$ have the same size. $\odot$ is an element-by-element multiplication. An illustration of $\mathbf{T}$ matrix is shown in Fig. 1. Then the output $\mathbf{O}$ represents the optimal aggregation of multiscale models $\{\mathbf{K}_l \otimes \mathbf{I}\mathbf{x}\}_{l=1}^{L}$. The scheme of our proposed DCL is shown in Fig. 2. The weight $\mathbf{T}$ is associated with the input and learned by an attention module, and the squeeze-and-excitation module [34] is introduced to calculate it. The global average pooling is first applied to squeeze the global spatial information of an input data cube. Two FCLs with a ReLU activation function between them are used to extract features. A specially designed layer, named Softmaxpro, is used to obtain sum-to-one attention weights $\mathbf{T}$ with the characteristics described above. To be specific, for the $\mathbf{T}_l$, a FCL is used to map the size of features the same as the lth convolutional kernel. Then a zero-padding operator makes it the same as the maximum size of convolutional kernels. Finally, we apply the Softmax to the available weights of each position of $\mathbf{T}$ to accomplish this goal

$$\{\mathbf{T}_l\}_{l=1}^{L}(i, j) = \text{Softmax}\big(\{\mathcal{T}(\mathbf{T}_l)\}_{l=1}^{L}(i, j)\big). \tag{27}$$

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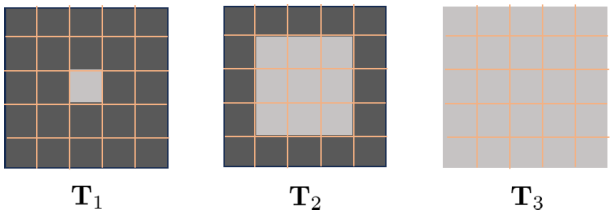


Fig. 1. Diagram of the $\mathbf{T}$ matrix. $L = 3$ for example with kernel sizes set to 1×1, 3×3, and 5×5. The black blocks represent the corresponding weights are zeros, the gray blocks are with available weights, and $\sum_{l=1}^{L} \mathbf{T}_l(i, j) = 1$ for the (i, j)th position.

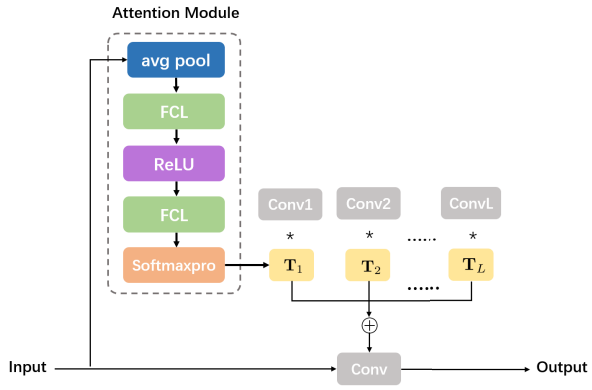


Fig. 2. Scheme of the DCL.

Our proposed DCL represents the optimal combination of conventional linear models. Compared with other multiscale unmixing methods [18], [27], [35], it does not require expanding the network depth and can effectively combine the multiscale spectral–spatial information.

With this useful and efficient DCL network component, (23) is equivalently reformulated as

$$\begin{aligned}\mathbf{A}^{(k+1)} &= \text{Layer}_A\left(\mathbf{V}_1^{(k)}, \mathbf{G}_1^{(k)}, \mathbf{X}; \{\mathbf{W}_1, \mathbf{Q}_1, \mathbf{R}, \mathbf{H}\}^{(k+1)}\right) \\ &= \sum_{l=1}^{L}\left(\mathcal{T}\left(\mathbf{R}_l^{(k+1)}\right) \odot \mathbf{W}_{1,l}^{(k+1)}\right) \otimes \mathbf{X} \\ &\quad + \sum_{p=1}^{P}\left(\mathcal{T}\left(\mathbf{H}_p^{(k+1)}\right) \odot \mathbf{Q}_{1,p}^{(k+1)}\right) \otimes \left(\mathbf{V}_1^{(k)} - \mathbf{G}_1^{(k)}\right) \quad (28)\end{aligned}$$

where L and P are the number of kernels, $\mathbf{W}_{1,l}$ and $\mathbf{Q}_{1,p}$ represent the weights of the lth and pth convolution kernels, and $\mathbf{R}_l$ and $\mathbf{H}_p$ are the attention weight matrices of the DCL.

Then we apply a Softmax operator to constrain the output of Layer$_A$ within $\mathcal{D}_A$, i.e.

$$\mathbf{A} = \text{Softmax}\left(\text{Layer}_A\left(\mathbf{V}_1^{(k)}, \mathbf{G}_1^{(k)}, \mathbf{X}; \{\mathbf{W}_1, \mathbf{Q}_1\}^{(k+1)}\right)\right). \quad (29)$$

The Layer$_A$ neural network structure is illustrated in Fig. 3.

2) Update $\mathbf{V}_1$*:* The V_1-subproblem is a standard RED objective function

$$\min_{\mathbf{V}_1} \frac{\alpha}{2}\|\mathbf{A} - \mathbf{V}_1 + \mathbf{G}_1\|_{\text{F}}^2 + \lambda \mathcal{R}_{\text{RED}}(\mathbf{V}_1). \quad (30)$$

It can also be solved in a closed form. We use the fixed-point strategy to solve this problem. By setting the gradient of the objective function to $\mathbf{0}$, we have the following equation:

$$\lambda\left(\mathbf{V}_1 - \mathsf{C}(\mathbf{V}_1)\right) + \alpha(\mathbf{A} - \mathbf{V}_1 + \mathbf{G}_1) = \mathbf{0}. \quad (31)$$

The solution of V_1-subproblem is an iterative scheme as follows:

$$\mathbf{V}_1^{(k+1)} = \frac{1}{\lambda + \alpha}\left[\lambda \mathsf{C}\left(\mathbf{V}_1^{(k)}\right) + \alpha\left(\mathbf{A}^{(k+1)} + \mathbf{G}_1^{(k)}\right)\right]. \quad (32)$$

The V_1-update layer is derived by unrolling (32). The update of the $(k+1)$th iteration of Layer$_{V_1}$ is expressed as

$$\begin{aligned}\mathbf{V}_1^{(k+1)} &= \text{Layer}_{V_1}\left(\mathbf{A}_1^{(k+1)}, \mathbf{G}_1^{(k)}; \theta_1^{(k+1)}, \theta_2^{(k+1)}\right) \\ &= \theta_1^{(k+1)}\mathsf{C}\left(\mathbf{V}_1^{(k)}\right) + \theta_2^{(k+1)}\left(\mathbf{A}^{(k+1)} + \mathbf{G}_1^{(k)}\right) \quad (33)\end{aligned}$$

where $\theta_1^{(k+1)}$ and $\theta_2^{(k+1)}$ are learnable parameters and play the role of $(\lambda/(\lambda+\alpha))$ and $(\alpha/(\lambda+\alpha))$. In this manner, we can use a data-driven strategy to learn the regularization parameters, rather than handcraft them. The Layer$_{V_1}$ neural network layer is shown in Fig. 3.

3) Update $\mathbf{M}$*:* As for the update of M-subproblem, it aims to solve the following optimization problem:

$$\min_{\mathbf{M}} \frac{1}{2}\|\mathbf{X} - \mathbf{M}\mathbf{A}\|_{\text{F}}^2 + \frac{\beta}{2}\|\mathbf{M} - \mathbf{V}_2 + \mathbf{G}_2\|_{\text{F}}^2 \quad (34)$$

which can be efficiently solved through a closed-form solution. Then we obtain the following update iteration:

$$\mathbf{M}^{(k+1)} = \left(\mathbf{X}\mathbf{A}^{\top} - \beta\mathbf{V}_2 + \beta\mathbf{G}_2\right)\left(\mathbf{A}\mathbf{A}^{\top} - \beta\mathbf{I}\right)^{-1}. \quad (35)$$

The M-update layer is designed by unfolding (35). With the $\mathbf{V}_2^{(k)}$, $\mathbf{G}_2^{(k)}$, and given $\mathbf{X}$, it is written as follows:

$$\begin{aligned}\mathbf{M}^{(k+1)} &= \text{Layer}_M\left(\mathbf{V}_2^{(k)}, \mathbf{G}_2^{(k)}, \mathbf{X}; \mathbf{W}_2, \mathbf{Q}_2\right) \\ &= \mathbf{X}\mathbf{W}_2 + \left(\mathbf{G}_2^{(k)} - \mathbf{V}_2^{(k)}\right)\mathbf{Q}_2 \quad (36)\end{aligned}$$

where

$$\begin{aligned}\mathbf{W}_2 &= \mathbf{A}^{\top}\left(\mathbf{A}\mathbf{A}^{\top} - \beta\mathbf{I}\right)^{-1} \\ \mathbf{Q}_2 &= \left(\mathbf{A}\mathbf{A}^{\top} - \beta\mathbf{I}\right)^{-1}\beta. \quad (37)\end{aligned}$$

We replace $\mathbf{W}_2$ and $\mathbf{Q}_2$ with learnable parameters to enhance flexibility. The Layer$_M$ neural network layer is presented in Fig. 3 and defined as

$$\mathbf{M}^{(k+1)} = \text{Layer}_M\left(\mathbf{V}_2^{(k)}, \mathbf{G}_2^{(k)}, \mathbf{X}; \{\mathbf{W}_2, \mathbf{Q}_2\}^{(k+1)}\right). \quad (38)$$

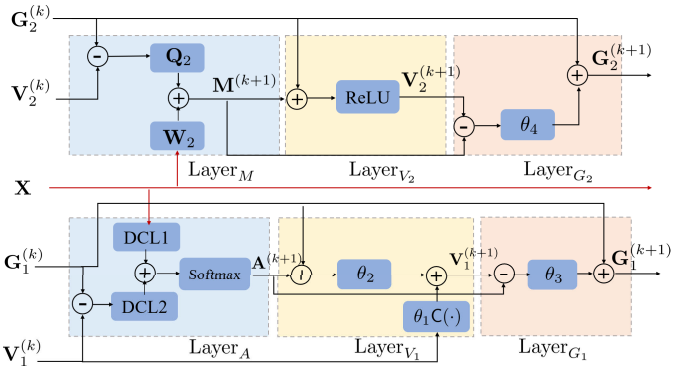


Fig. 3. Architecture of an unrolling layer (one block).

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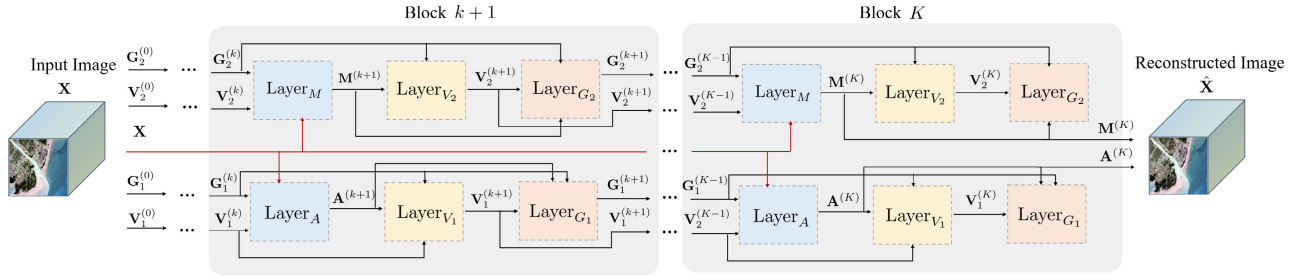


Fig. 4. Architecture of the proposed PnP-Net with K blocks.

4) Update $\mathbf{V}_2$: The V_2-subproblem is a least-square problem with the nonnegative constraint, which is formulated as

$$\min_{\mathbf{V}_2} \frac{\beta}{2}\|\mathbf{M}-\mathbf{V}_2+\mathbf{G}_2\|_{\mathrm{F}}^2+\Omega_{\mathcal{D}_M}(\mathbf{V}_2). \tag{39}$$

We use a hard threshold operator to solve this problem by mapping $\mathbf{M}+\mathbf{G}_2$ to a set of all nonnegative elements. This operation is written as

$$\mathbf{V}_2=\max(\mathbf{M}+\mathbf{G}_2,\mathbf{0}). \tag{40}$$

In the design of Layer$_{V_2}$, we use the ReLU to generate this layer

$$\begin{aligned}\mathbf{V}_2^{(k+1)}&=\mathrm{Layer}_{V_2}\left(\mathbf{M}^{(k+1)},\mathbf{G}_2^{(k)}\right)\\&=\mathrm{ReLU}\left(\mathbf{M}^{(k+1)}+\mathbf{G}_2^{(k)}\right).\end{aligned} \tag{41}$$

The Layer$_{V_2}$ neural network layer is shown in Fig. 3.

5) Update $\mathbf{G}_1$ and $\mathbf{G}_2$: The Layer$_{G_1}$ is designed to unfold the following iteration:

$$\mathbf{G}_1=\mathbf{G}_1+\eta_1(\mathbf{A}-\mathbf{V}_1). \tag{42}$$

With $\mathbf{A}^{(k+1)}$ and $\mathbf{V}_1^{(k+1)}$, the $(k+1)$th neural network layer to update $\mathbf{G}_1$ is designed as

$$\begin{aligned}\mathbf{G}_1^{(k+1)}&=\mathrm{Layer}_{G_1}\left(\mathbf{A}^{(k+1)},\mathbf{V}_1^{(k+1)};\theta_3^{(k+1)}\right)\\&=\mathbf{G}_1^{(k)}+\theta_3^{(k+1)}\left(\mathbf{A}^{(k+1)}-\mathbf{V}_1^{(k+1)}\right)\end{aligned} \tag{43}$$

where $\theta_3^{(k+1)}$ is a learnable parameter to replace the role of η_1.

Layer$_{G_2}$ is derived from

$$\mathbf{G}_2=\mathbf{G}_2+\eta_2(\mathbf{M}-\mathbf{V}_2). \tag{44}$$

The same as the design of Layer$_{G_1}$, this layer uses $\theta_4^{(k+1)}$ to replace the role of η_2

$$\begin{aligned}\mathbf{G}_2^{(k+1)}&=\mathrm{Layer}_{G_2}\left(\mathbf{M}^{(k+1)},\mathbf{V}_2^{(k+1)};\theta_4^{(k+1)}\right)\\&=\mathbf{G}_2^{(k)}+\theta_4^{(k+1)}\left(\mathbf{M}^{(k+1)}-\mathbf{V}_2^{(k+1)}\right).\end{aligned} \tag{45}$$

The Layer$_{G_1}$ and Layer$_{G_2}$ neural network layers are also shown in Fig. 3.

C. Network Architecture

Each unrolling block represents a single iteration of the ADMM-based plug-and-play unmixing method and consists of six parts: Layer$_A$, Layer$_{V_1}$, Layer$_{G_1}$, Layer$_M$, Layer$_{V_2}$, and Layer$_{G_2}$. As shown in Fig. 4, we use K iteration blocks to conduct the architecture of our proposed method, which can mimic the ADMM-based unmixing algorithm with K iterations. For the kth block, the learnable parameters are $\mathbf{\Theta}^{(k)}=\{\mathbf{R}^{(k)},\mathbf{W}_1^{(k)},\mathbf{H}^{(k)},\mathbf{Q}_1^{(k)},\theta_1^{(k)},\theta_2^{(k)},\mathbf{W}_2^{(k)},\mathbf{Q}_2^{(k)},\theta_3^{(k)},\theta_4^{(k)}\}$. Each block in the network has its own set of learnable parameters that are not shared across blocks, which has demonstrated flexibility and strong learning capability for unmixing tasks.

Additionally, the PnP-Net architecture can be divided into two parts: the endmember estimation network and the abundance estimation network. Each block of the endmember estimation network contains the Layer$_M$, Layer$_{V_2}$, and Layer$_{G_2}$ components, there are K blocks to estimate the endmember. The abundance estimation network comprises three parts, namely Layer$_A$, Layer$_{V_1}$, and Layer$_{G_1}$. K layers are applied to estimate the abundance.

D. Training Details and Network Initialization

Our PnP-Net is trained to jointly estimate endmember and abundance with the input hyperspectral image $\mathbf{X}$. We use the mean square error (MSE) to compute the differences between the input and its reconstruction. Compared to calculating the MSE loss directly using the reconstruction of the final Kth block output, calculating it for each block can better constrain the training process of the network. Thus, we use the weighted sum of the reconstructions from each block to compute the loss function, which is defined as

$$\mathcal{L}=\frac{1}{2N}\sum_{k=1}^{K}\beta_k\left\|\mathbf{X}-\hat{\mathbf{X}}^{(k)}\right\|_{\mathrm{F}}^2 \tag{46}$$

where β_k denotes the importance of each term, and $\hat{\mathbf{X}}^{(k)}=\hat{\mathbf{M}}^{(k)}\hat{\mathbf{A}}^{(k)}$ is the corresponding reconstructed image of the kth block.

Parameter initialization plays an important role in our PnP-Net, which can accelerate the convergence speed and enhance the endmember and abundance estimation accuracy. The endmember matrix $\mathbf{M}^{(0)}$ and abundance matrix $\mathbf{A}^{(0)}$ are calculated by VCA [36] and FCLS [37]. Specially, we obtain the initialization of $\{\mathbf{W}_1,\mathbf{Q}_1,\mathbf{W}_2,\mathbf{Q}_2\}$ according to $\mathbf{A}^{(0)}$ and $\mathbf{M}^{(0)}$, and other values are initialized with zeros.

Our PnP-Net method is flexible to plug various deep denoisers. In this work, we choose three novel denoisers, i.e., DnCNN [38], IRCNN [39] and SCUNet [40], as examples. We pretrain these models using images of Waterloo Exploration Database [41], DIV2K [42], and Flick2K [43] datasets with different noise settings. It is worth noting that the parameters of denoiser are fixed during training which can reduce computational stress and take advantage of outer priors.

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Our PnP-Net with these denoisers are named PnP-Net-1, PnP-Net-2, and PnP-Net-3. During training, we set the learning rate to 5×10^{-4}. The number of iterations blocks (K) is set to 5. L and P are set to 3 with the kernel size setting to 1×1, 3×3, and 5×5. We set $\{\beta_k\}_{k=1}^{K}$ to $\{1 \times 10^{-4}, 1 \times 10^{-3}, 1 \times 10^{-2}$, and $1 \times 10^{-1}, 1\}$.

IV. Experiment Results

In this section, we first introduce the evaluation indicators to analyze the effectiveness of the proposed methods. The considered conventional and state-of-the-art compared methods are also briefly presented. Then, the datasets used in the experiments and results are illustrated and discussed.

A. Evaluation Indicators

We use several indicators to evaluate the unmixing results quantitatively in terms of estimated abundances, extracted endmembers, and reconstructed images. In addition, we have no available ground-truth for the real datasets, we only evaluate the reconstructed image quantitatively and subjectively evaluate the visual effect.

The root mean square error (RMSE) is applied to assess the differences between the ground-truth and estimated abundances and endmembers. We respectively name them aRMSE and mRMSE, which are defined as follows:

$$\text{aRMSE} = \sqrt{\frac{1}{NR}\sum_{i=1}^{N}\left\|\mathbf{a}_i - \hat{\mathbf{a}}_i\right\|^2} \tag{47}$$

and

$$\text{mRMSE} = \sqrt{\frac{1}{BR}\sum_{r=1}^{R}\left\|\mathbf{m}_r - \hat{\mathbf{m}}_r\right\|^2} \tag{48}$$

where $\hat{\mathbf{a}}_i$ and $\mathbf{a}_i$ represent the estimated and real abundances of the ith pixel, and $\hat{\mathbf{m}}_r$ and $\mathbf{m}_r$ denote the rth extracted and corresponding real endmembers.

We apply the mean spectral angle distance (SAD) to evaluate the degree of distortion between the real and reconstructed spectral signatures, and mSAD denotes the SAD between the estimated and real endmembers. They are calculated by

$$\text{SAD} = \frac{1}{N}\sum_{i=1}^{N}\arccos\left(\frac{\mathbf{y}_i^{\top}\hat{\mathbf{y}}_i}{\|\mathbf{y}_i\|\left\|\hat{\mathbf{y}}_i\right\|}\right) \tag{49}$$

and

$$\text{mSAD} = \frac{1}{R}\sum_{r=1}^{R}\arccos\left(\frac{\mathbf{m}_i^{\top}\hat{\mathbf{m}}_r}{\|\mathbf{m}_r\|\left\|\hat{\mathbf{m}}_r\right\|}\right) \tag{50}$$

where $\mathbf{y}_i$ and $\hat{\mathbf{y}}_i$ represent the real and reconstructed spectra.

We also use the peak signal-to-noise ratio (PSNR) to evaluate the differences between the real and reconstructed images, calculated by

$$\text{PSNR} = 10 \times \log_{10}\left(\frac{\max^2}{\text{mse}}\right) \tag{51}$$

where max represents the highest pixel value in the reconstructed image.

B. Compared Methods

We compare the proposed PnP-Net method with several conventional and state-of-the-art unmixing methods, i.e., PnP-NMF [30], gtvMBO [9], U-ADMM-BUNet [26], DIFCNN [27], and AERED [44]. To make a fair comparison, all methods are initialized by the endmembers extracted by VCA [36] and abundances estimated by FCLS [37]. All parameters of these methods are carefully chosen to achieve the best experimental results.

The first two compared methods are conventional. The PnP-NMF is a classic plug-and-play based unmixing method, we use SCUNet as the deep denoiser to plug priors. The gtvMBO method is a joint endmember and abundance estimation-based unmixing method with a data-driven graph TV regularization, and its objective function is solved by ADMM. The U-ADMM-BUNet and DIFCNN are novel unrolling-based unmixing methods. AERED integrates the autoencoder-based unmixing network with RED, which combines the explicit and implicit priors. We name the AERED with NLM denoiser and block-matching and 4-D (BM4D) filtering [45] denoiser as AERED-1 and AERED-2, respectively.

C. Experiments on Synthetic Datasets

To quantitatively verify the unmixing results, we generate a set of synthetic data with the spatial size of 100×100. We adapt the method described in [46] to produce these data. The ground truth of abundance maps is shown in Fig. 5, and they are constrained to satisfy ASC and ANC. The abundance obtained by this strategy shows a spatial correlation that is similar to the real data. Four spectral signatures selected from the USGS spectral library with 224 bands are applied to generate the endmember matrix. The reference endmember spectra are shown in Fig. 6. Zero mean Gaussian noises with four levels of signal-to-noise-ratio (SNR), i.e., 5, 10, 20, and 30 dB, are added to the data.

Table I shows the aRMSE and PSNR results of the synthetic data, and Table II lists the mRMSE and mSAD results. The bold numbers in Tables I and II denote the best results, and the underlined numbers indicate the second best results. From the results of these two tables, it can be observed that the proposed PnP-Net unmixing framework achieves the best unmixing results compared to the benchmark methods. Compared to traditional unmixing methods such as PnP-NMF and gtvMBO, the proposed method eliminates the need for penalty parameter selection, instead utilizing a data-driven learning approach. In comparison to the unrolling unmixing method, i.e., U-ADMM-BUNet and DIFCNN, this proposed PnP-Net method employs dynamic convolution to achieve multiscale information learning and fusion and gets better results. Our PnP-Net method leverages an externally pretrained denoiser on large datasets to provide outer priors, and also integrates a hyperspectral data-driven and physically interpretable neural network to learn internal priors. This combination results in superior unmixing performance. We can also observe that, due to the use of deep denoisers and DNNs, our proposed unmixing method is robust to noise.

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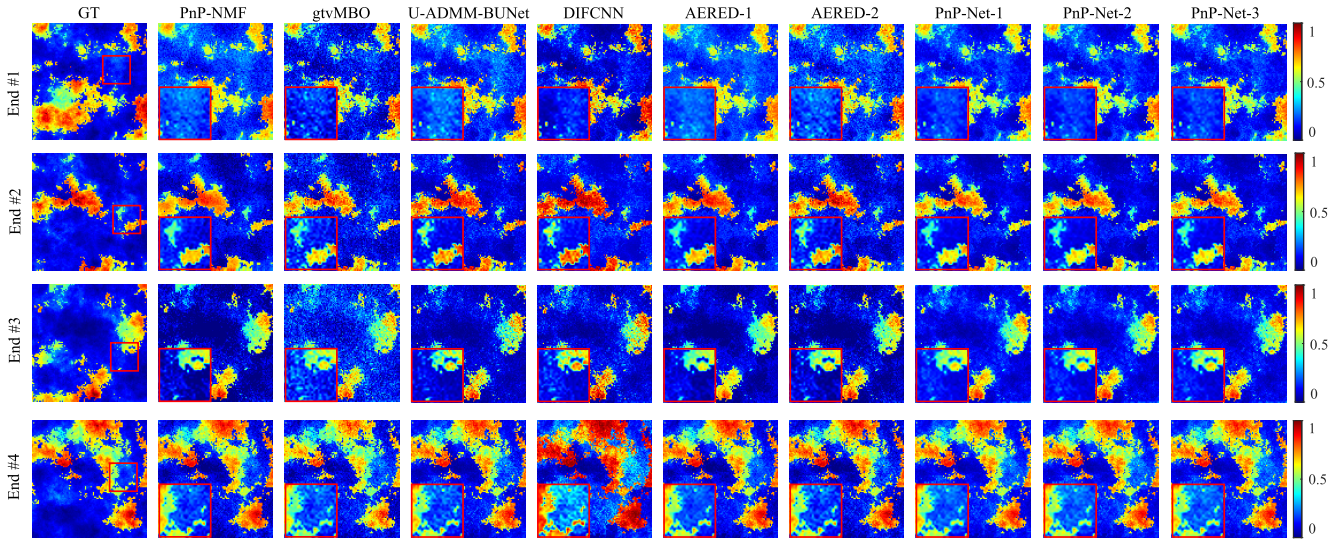


Fig. 5. Illustration of the ground-truth and estimated abundance maps (SNR = 10 dB). From top to bottom different endmembers. From left to right different compared methods. The subfigures zoomed-out view the region within the red box on the ground-truth map.

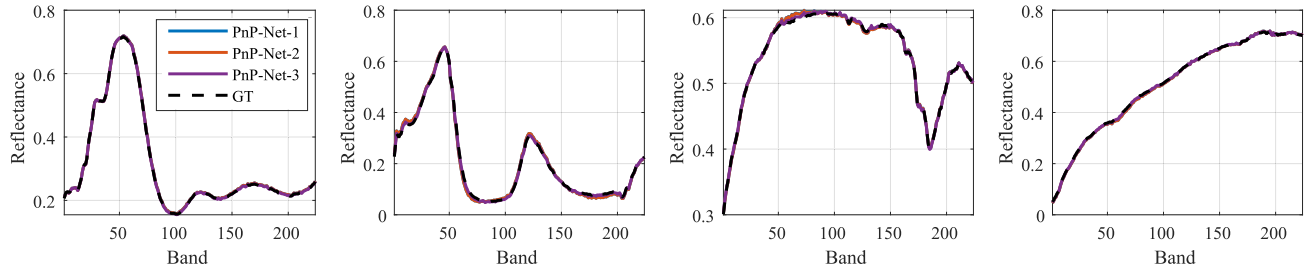


Fig. 6. Illustration of the ground-truth and estimated endmembers (SNR = 20 dB).

TABLE I

PERFORMANCE EVALUATION OF THE SYNTHETIC DATA (IN TERMS OF aRMSE $\times 10^{-2}$ AND PSNR)

Noise level		5dB		10dB		20dB		30dB	
Method	Denoiser	aRMSE ↓	PSNR ↑	aRMSE ↓	PSNR ↑	aRMSE ↓	PSNR ↑	aRMSE ↓	PSNR ↑
PnP-NMF	/	9.05±0.28	32.53±1.73	6.11±0.17	36.92±0.94	2.23±0.10	44.12±0.47	0.80±0.01	55.21±0.39
gtvMBO	/	11.49±0.31	29.61±2.03	6.99±0.21	34.30±0.98	2.72±0.10	43.99±0.52	0.80±0.03	53.86±0.32
U-ADMM-BUNet	/	9.39±0.29	30.67±1.58	6.47±0.19	35.14±0.64	2.56±0.13	44.00±0.49	0.86±0.03	53.37±0.33
DIFCNN	/	10.44±0.33	30.48±1.71	6.51±0.21	34.88±0.75	2.42±0.12	44.07±0.47	0.81±0.02	53.16±0.37
AERED	NLM	9.07±0.21	32.22±1.57	6.04±0.19	35.62±0.63	2.46±0.10	44.13±0.40	0.81±0.01	54.48±0.28
	BM3D	9.27±0.23	32.47±1.29	5.97±0.17	36.36±0.54	2.17±0.09	44.14±0.41	0.80±0.01	54.63±0.31
PnP-Net	DnCNN	8.56±0.24	**33.51±1.46**	5.72±0.19	37.61±0.65	**1.97±0.08**	**44.35±0.36**	**0.79±0.01**	**55.60±0.34**
	IRCNN	**8.50±0.20**	33.43±1.36	5.66±0.18	**37.73±0.78**	2.03±0.09	44.16±0.46	0.80±0.01	55.47±0.37
	SCUNet	8.59±0.21	33.31±1.28	**5.35±0.16**	**37.73±0.88**	1.99±0.06	44.29±0.45	0.80±0.01	55.48±0.39

Fig. 5 shows the abundance maps of the proposed and competing methods for the synthetic data at 10 dB. We observe that all these estimates are very close to the ground truth. Moreover, it can be seen that the noise in the abundance maps of PnP-Net and AERED is obviously smaller than that of other comparison methods, which indicates the superiority of combination deep networks with denoisers. Furthermore, Fig. 6 illustrates the extracted endmembers of the proposed method. We observe that the result of our method is close to the ground truth. Fig. 7 illustrates how the number of blocks affects the unmixing performance and also shows how the number of kernels L and P impact the unmixing results. We can observe that within an optimal range of settings, our method can obtain good results.

D. Experiments on Real Datasets

We also evaluate our proposed method on real datasets. We conduct experiments on two widely used hyperspectral images obtained by airborne sensors, namely the Jasper Ridge dataset and the Muufl Gulfport dataset. The descriptions and experimental results of these two datasets are presented below.

The Jasper Ridge dataset was captured by Analytical Imaging and Geophysics (AIG) in 1999. It contains 224 bands

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TABLE II
PERFORMANCE EVALUATION OF THE SYNTHETIC DATA (IN TERMS OF MRMSE $\times 10^{-2}$ AND MSAD)

Noise level		5dB		10dB		20dB		30dB	
Method	Denoiser	mRMSE ↓	mSAD ↓	mRMSE ↓	mSAD ↓	mRMSE ↓	mSAD ↓	mRMSE ↓	mSAD ↓
PnP-NMF	/	4.09±0.45	6.28±0.56	2.03±0.13	2.80±0.27	0.52±0.01	0.93±0.11	**0.13±0.00**	**0.19±0.01**
gtvMBO	/	5.02±0.51	6.41±0.38	2.05±0.14	2.78± 0.25	1.00±0.01	1.10±0.10	0.14±0.00	**0.19±0.03**
U-ADMM-BUNet	/	4.00±0.49	6.31±0.39	2.00±0.19	3.25±0.32	0.50±0.01	0.95±0.17	0.15±0.00	0.20±0.03
DIFCNN	/	4.23±0.72	6.42±0.41	1.99±0.17	3.03±0.29	0.67±0.02	1.11±0.15	0.17±0.00	0.20±0.04
AERED	NLM	4.15±0.63	6.27±0.40	2.04±0.16	2.91±0.25	0.51±0.01	1.01±0.11	0.15±0.00	0.21±0.02
	BM3D	4.01±0.48	6.29±0.35	2.06±0.10	2.76±0.19	0.49±0.01	0.97±0.09	0.14±0.00	0.21±0.01
PnP-Net	DnCNN	3.94±0.44	6.22±0.38	1.94±0.09	2.72±0.16	0.46±0.01	**0.87±0.08**	**0.13±0.00**	**0.19±0.01**
	IRCNN	3.99±0.42	6.29±0.39	2.02±0.11	2.78±0.20	0.49±0.01	0.93±0.12	0.14±0.00	**0.19±0.01**
	SCUNet	**3.90±0.41**	**6.13±0.37**	**1.92±0.08**	**2.65±0.17**	**0.42±0.01**	**0.87±0.09**	0.14±0.00	**0.19±0.01**

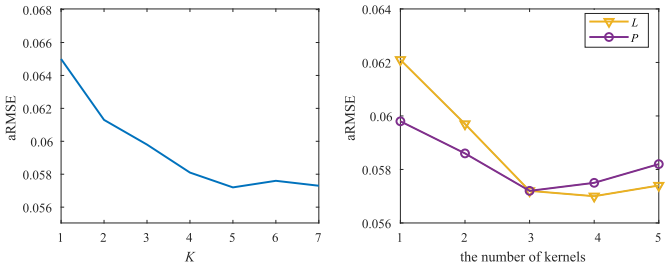


Fig. 7. Left figure shows the impact of the number of iterations blocks K on abundance estimation, and the right figure shows the impact of the number of kernels L and P on abundance estimation.

covering the spectral range from 380 to 2500 nm. After removing the bands affected by water vapor and atmospheric (1–3, 108–112, 154–166, and 220–224), 198 bands are remained. The original size of this data is 512 × 614, a popular region of interest with 100 × 100 pixels cropped. The four endmembers in this data are "water," "soil," "tree," and "road."

The Muufl Gulfport dataset was captured by the CASI-1500 hyperspectral sensor over the University of Southern Mississippi Gulf Park Campus. The original image contains 72 bands with 325 × 220 pixels. We remove the first four and the last four bands of the image, which contain a lot of noise. A subimage with the size of 130 × 90 pixels is applied in our experiments. We reference the scene label ground-truth map from manually labeling in [47]. There are five pure materials in this subimage, namely "roof," "grass," "tree," "shadow," and "asphalt."

The visual results of the estimated abundance maps of these two datasets are shown in Figs. 8 and 9. The first four rows of Fig. 8 report the comparisons of our and compared methods with respect to the Jasper Ridge dataset. All these methods decompose this data with four clear abundance maps. Our PnP-Net achieves clearer abundance results. The last line of Fig. 8 illustrates the reconstructed error between the reconstructed and real images of these methods. We observe that our method gets good reconstructed results which are also consistent with the results in Table III. To some extent, it proves the advantages of our method to unmix the real data. The abundance maps of the Muufl Gulfport dataset are illustrated in Fig. 9. It can be seen that the results of gtvMBO are more noisy than other methods, especially for the "asphalt" endmember. The proposed PnP-Net method can estimate the abundance maps much clearer than the compared method. Our results also have more detailed spatial information and sharper edges. The last row of Fig. 9 presents the reconstructed error maps of all methods. DIFCNN shows higher results. It may be caused by the loss function selected to train the model. It uses the cross entropy-loss to train the network, while other compared deep methods apply the MSE-like loss to train the model. The latter is more conductive to reduce the Euclidean distance between the reconstructed and original images. Table IV lists the quantitative results of the Muufl Gulfport data. We get that our method can obtain good reconstructions.

E. Method Analysis

1) Fine-Tuning the Denoiser: The parameters of plugged deep denoisers can also be fine-tuned during the network training. We set the fine-tuning learning rate to 1×10^{-5}. These methods are respectively named U-DnCNN, U-IRCNN, and U-SCUNet. The unmixing results are listed in Tables V and VI. Compared with the results in Tables I and II, we can observe that fine-tuning the parameters of the deep denoiser can improve the denoising network's generalization ability for hyperspectral data and obtain slightly better unmixing performance, but it significantly increases the computational load and demands more computing resources [Parameter comparison: 0.027M (PnP-Net) versus 0.693M (U-DnCNN) versus 0.213M (U-IRCNN) versus 17.971M (U-SCUNet)].

2) Ablation Studies: We conduct ablation studies to evaluate the effect of each block. First, we compare our proposed DCLs with four different network layers: FCL, original convolution layer (CNN-1), convolution layer with multiple kernel sizes for different iterative layers (CNN-M), and dynamic convolution with the same kernel size (PnP-Net-1). We set the kernel size of CNN-1 to 3 × 3. CNN-M is designed with the kernel size set to 1 × 1 for the first three iteration blocks and 3 × 3 for the last two iteration blocks. The kernel sizes of PnP-Net-1 are set to 3 × 3. Second, we perform ablation studies to assess the impact of employing RED. The proposed method without RED regularization, but directly using the deep denoiser is instantiated as PnP-Net-2, and without using

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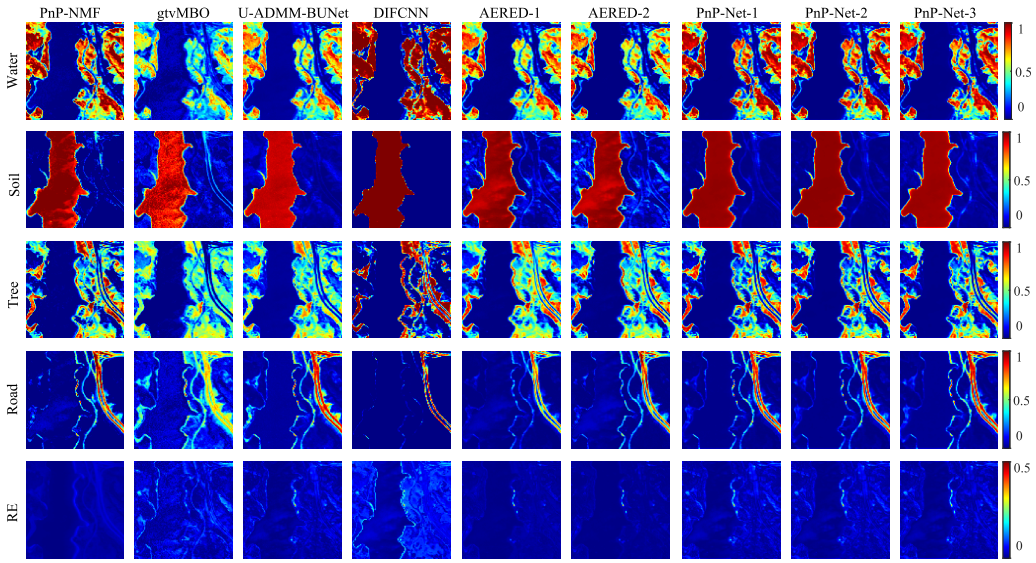


Fig. 8. Illustration of the results of compared methods and proposed PnP-Net on Jasper Ridge dataset. (The first four rows are estimated abundances of different endmembers, and the last row is the map of reconstructed error.)

TABLE III

RECONSTRUCTION PERFORMANCE EVALUATION OF THE JASPER RIDGE DATA (IN TERMS OF PSNR AND SAD)

Method	PnP-NMF	gtvMBO	U-ADMM-BUNet	DIFCNN	AERED		PnP-Net		
Denoiser	/	/	/	/	NLM	BM3D	DnCNN	IRCNN	SCUNet
PSNR ↑	**32.34±1.99**	29.59±1.87	30.01±1.12	29.73±1.94	31.39±1.17	30.04±1.46	29.91±1.47	30.12±1.61	30.08±1.57
SAD ↓	6.30±1.00	6.32±1.02	5.64±0.95	8.76±1.52	5.55±0.94	5.44±0.98	5.54±0.96	**5.42±1.03**	5.62±0.97

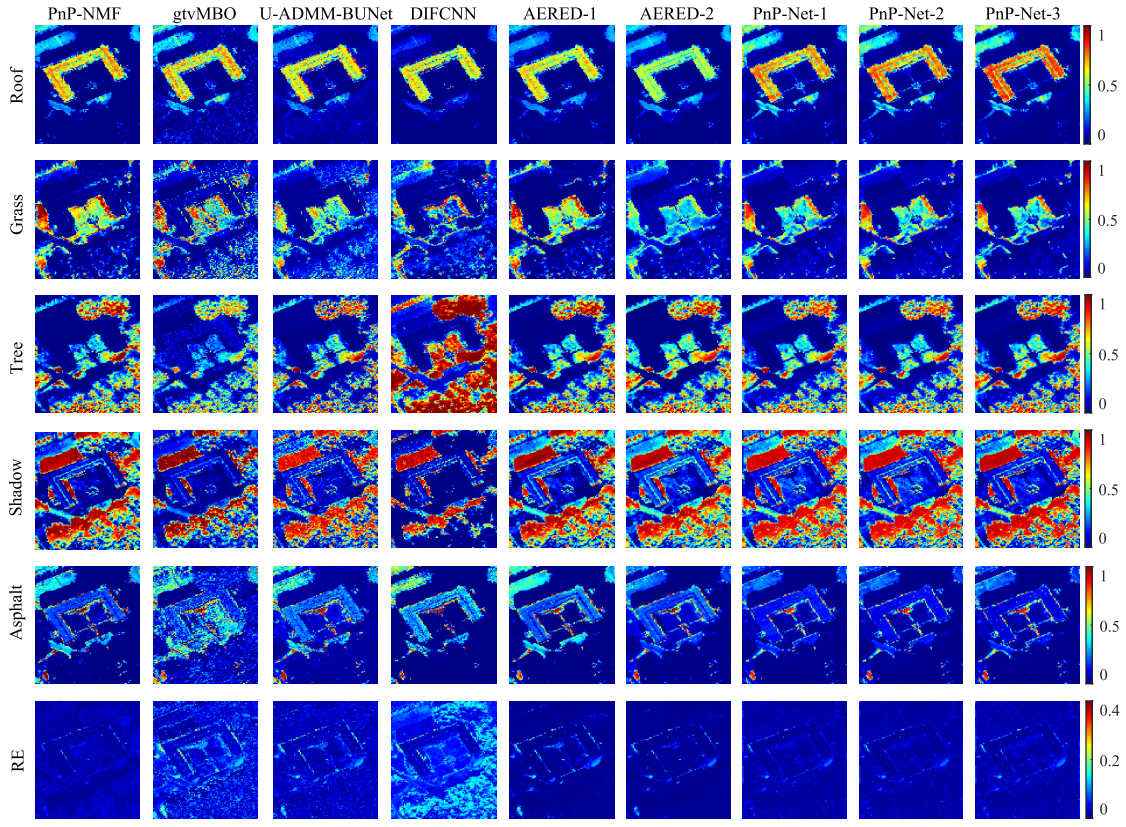


Fig. 9. Illustration of the results of compared methods and our PnP-Net on Muufl Gulfport dataset. (The first five rows are estimated abundances of different endmembers, and the last row is the map of reconstructed error.)

deep denoiser is named PnP-Net-3. The synthetic data with SNR = 10 dB are applied to perform the experiments with the DnCNN denoiser as an example, the aRMSE results of the estimated abundances are shown in Table VII. By comparing

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TABLE IV

RECONSTRUCTION PERFORMANCE EVALUATION OF THE MUUFL GULFPORT DATA (IN TERMS OF PSNR AND MSAD)

Method	PnP-NMF	gtvMBO	U-ADMM-BUNet	DIFCNN	AERED		PnP-Net		
Denoiser	/	/	/	/	NLM	BM3D	DnCNN	IRCNN	SCUNet
PSNR ↑	32.15±0.97	30.17±0.92	31.15±0.88	25.65±0.95	32.32±1.17	**33.44±1.07**	32.48±1.20	32.55±1.28	32.34±1.17
SAD ↓	5.24±1.02	7.58±1.05	6.86±0.92	8.39±1.26	5.10±0.84	**4.48±0.86**	4.99±0.96	4.95±0.93	5.10±0.95

TABLE V

ARMSE $\times 10^{-2}$ AND PSNR RESULTS OF THE SYNTHETIC DATASET WITH THE DENOISER PARAMETER FINE-TUNING

	U-DnCNN		U-IRCNN		U-SCUNet	
	aRMSE	PSNR	aRMSE	PSNR	aRMSE	PSNR
5dB	8.53 ± 0.25	33.67 ± 1.77	8.46 ± 0.20	33.52 ± 1.37	8.41 ± 0.19	33.72 ± 1.33
10dB	5.67 ± 0.18	37.87 ± 0.81	5.61 ± 0.15	37.71 ± 0.69	5.26 ± 0.17	37.76 ± 0.71
20dB	1.94 ± 0.07	44.38 ± 0.39	1.93 ± 0.06	44.59 ± 0.41	1.91 ± 0.06	44.41 ± 0.44
30dB	0.80 ± 0.01	55.20 ± 0.41	0.76 ± 0.01	55.51 ± 0.39	0.82 ± 0.01	55.47 ± 0.37

TABLE VI

MRMSE $\times 10^{-2}$ AND MSAD RESULTS OF THE SYNTHETIC DATASET WITH THE DENOISER PARAMETER FINE-TUNING

	U-DnCNN		U-IRCNN		U-SCUNet	
	mRMSE	mSAD	mRMSE	mSAD	mRMSE	mSAD
5dB	3.91 ± 0.41	6.29 ± 0.39	3.93 ± 0.45	6.28 ± 0.40	3.89 ± 0.39	6.26 ± 0.41
10dB	1.92 ± 0.09	2.50 ± 0.14	1.99 ± 0.10	2.63 ± 0.18	1.94 ± 0.08	2.57 ± 0.16
20dB	0.47 ± 0.01	0.88 ± 0.09	0.48 ± 0.01	0.92 ± 0.10	0.49 ± 0.01	0.86 ± 0.08
30dB	0.13 ± 0.00	0.20 ± 0.01	0.15 ± 0.00	0.20 ± 0.01	0.14 ± 0.00	0.20 ± 0.01

TABLE VII

ARMSE RESULTS OF ABLATION STUDY FOR DIFFERENT NETWORK ARCHITECTURES WITH SYNTHETIC DATA (SNR = 10 dB)

FCL	CNN-1	CNN-M	PnP-Net-1	PnP-Net-2	PnP-Net-3
0.0598	0.0583	0.0576	0.0578	0.0577	0.0601

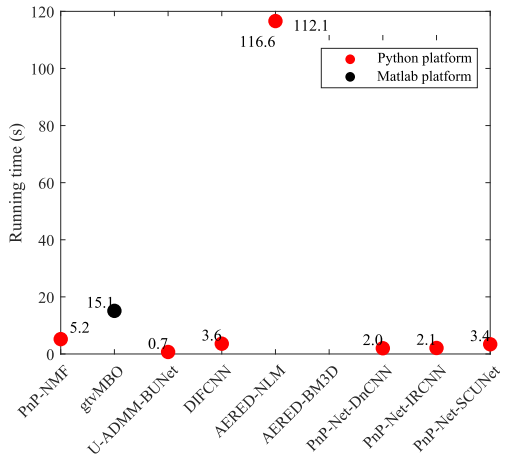


Fig. 10. Running time comparison.

the experimental results of different architectures, our method achieves the best performance, demonstrating the effectiveness of our proposed structure.

3) Running Time: We conduct experiments to evaluate the running time. All the experiments are conducted on a server with an AMD EPYC 7402 24-Core CPU and an NVIDIA GeForce RTX 3090 GPU. The time consumption of synthetic data is shown in Fig. 10. From the results, we can see that our proposed method is efficient compared with other methods.

V. CONCLUSION

In this article, we propose a novel PnP-Net method for hyperspectral unmixing. We unroll the plug-and-play unmixing method into a trainable deep model. We apply the RED to add denoising priors, and a various of denoisers can be plugged. The ADMM is used to solve the optimization function. We unfold the ADMM-based iterative steps to design the layer, and one layer represents an iteration. K layers are stacked to conduct the deep architecture. Through unrolling, we can use the end-to-end manner to train the network. To fully exploit the spatial information, we use the dynamic convolution with different sizes of convolution kernels to capture multiscale information. The denoisers are pretrained with a variety of images which makes our method leverage the power of external information to enhance the modeling ability. This strategy bypasses the issue of limited hyperspectral data and provides a way to integrate the inner and outer priors. Extensive experiment results show the superior performance of our method. In future work, we plan to plug generative models in our framework to further enhance the unmixing accuracy and exploit automatically adjusting network structure.

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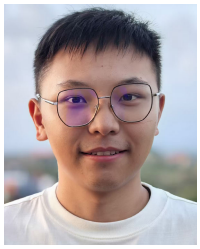
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Min Zhao (Graduate Student Member, IEEE) received the B.S. degree in electronic and information engineering from Northwestern Polytechnical University, Xi'an, China, in 2017, and the M.S. and Ph.D. degrees in signal and information processing from Northwestern Polytechnical University, in 2020 and 2024, respectively.

She is currently affiliated with the University of Hong Kong, Hong Kong. Her research interests include hyperspectral image analysis and object detection.

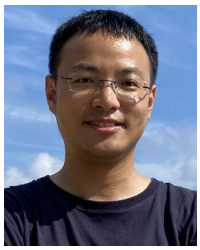
Ms. Zhao received the People's Choice Award of Three Minute Thesis (3MT) Competition of IEEE IGARSS 2020, and the (with team) Champion of Grand Challenges on NIR Image Colorization of IEEE VCIP 2020.



Linruize Tang (Graduate Student Member, IEEE) received the B.S. degree in information engineering from Northwestern Polytechnical University, Xi'an, China, in 2020, where he is currently pursuing the Ph.D. degree with the Centre for Intelligent Acoustics and Immersive Communications, School of Marine Science and Technology.

His research interests include hyperspectral image analysis, image fusion, and object detection.

Mr. Tang received the Runner-Up for the Best Student Paper at IEEE MLSP 2022.



Jie Chen (Senior Member, IEEE) received the B.S. degree from Xi'an Jiaotong University, Xi'an, China, in 2006, the Dipl.-Ing. degree in information and telecommunication engineering from the University of Technology of Troyes (UTT), Troyes, France, the M.S. degree in information and telecommunication engineering from Xi'an Jiaotong University, in 2009, and the Ph.D. degree in systems optimization and security from UTT, in 2013.

From 2013 to 2014, he was with the Lagrange Laboratory, University of Nice Sophia Antipolis, Nice, France. From 2014 to 2015, he was with the Department of Electrical Engineering and Computer Science, University of Michigan, Ann Arbor, MI, USA. He is a Professor with Northwestern Polytechnical University, Xi'an. His research interests include adaptive signal processing and distributed optimization, with application to image and acoustic processing.

Dr. Chen served as a Distinguished Lecturer for Asia-Pacific Signal and Information Processing Association from 2018 to 2019, the Co-Chair for the IEEE Signal Processing Society Summer School 2019 in Xi'an, the General Co-Chair for IEEE MLSP 2022, IEEE ICSPCC 2023 and 2024, and the Technical Co-Chair of IWAENC 2016 and IEEE ICSPCC 2022.

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